

JOB PORTAL MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

By

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Full Stack Application Development

Slot : E1



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May, 2026

BONAFIDE CERTIFICATE

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(K.Sai Manikanta)

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ABSTRACT

The Job Portal Management System is a web-based application designed to bridge the gap between job seekers and employers by providing a centralized platform for recruitment activities. The system enables students or job seekers to create profiles, upload resumes, search for job opportunities, and apply for positions based on their skills, experience, and preferences. At the same time, employers can post job openings, manage listings, review applications, and shortlist suitable candidates efficiently. On the frontend, the system uses HTML, CSS, JavaScript, and Thymeleaf to create a responsive and user-friendly interface. Additional features such as resume file upload, email notifications, and application status tracking enhance user experience. Advanced functionalities like OTP verification, chatbot integration, and analytics dashboards can be incorporated to improve system efficiency and usability. To enhance security, the system implements Spring Security with JWT (JSON Web Token) for authentication and role-based authorization, ensuring secure access for different types of users such as students, employers, and administrators. Additional features such as OTP verification, email notifications, resume upload functionality, chatbot support, and map integration further improve the usability and effectiveness of the system.

Keywords: *Job Portal, Recruitment System, Spring Boot, REST API, Role-Based Authentication, Resume Management, MySQL Database.*

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LIST OF ACRONYMS AND ABBREVIATIONS

JPA	Java Persistence API
ORM	Object Relational Mapping
REST	Representational State Transfer
DTO	Data Transfer Object
HTML	HyperText Markup Language
AJAX	Asynchronous JavaScript and XML
JSON	JavaScript Object Notation
SMTP	Simple Mail Transfer Protocol
IDE	Integrated Development Environment
DTO	Data Transfer Object
SQL	Structured Query Language
UI	User Interface
UX	User Experience
API	Gateway Application Programming Interface Gateway

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Chapter 1

INTRODUCTION

1.1 Introduction

The Job Portal Management System is a web-based application developed to simplify and modernize the recruitment process by providing a centralized platform for job seekers and employers.. Job portals play a crucial role in bridging the gap between job seekers and employ-ers by providing a centralized system for job searching, application submission, and recruitment management. The Job Portal Management System is designed to simplify and automate the hiring process through an efficient and user-friendly web-based platform. This sys-tem allows job seekers to create profiles, upload resumes, and apply for jobs seamlessly, while employers can post job vacancies and man-age applications effectively. By integrating modern web technologies, the platform ensures accessibility, scalability, and enhanced user experience. The Job Portal Manage- ment System is designed to simplify and automate the hiring process through an efficient and user-friendly web-based platform. This sys- tem allows job seekers to create profiles, upload resumes, and apply for jobs seamlessly, while employers can post job vacancies and man- age applications effectively. By integrating modern web technologies, the platform ensures accessibility, scalability, and enhanced user expe- rience. Features such as secure authentication, OTP-based email veri- fication, and real-time updates make the system reliable and efficient.

1.2 Aim of the Project

The primary aim of the Job Portal Management System is to design and develop a comprehensive web-based platform that facilitates efficient interaction between job seekers and employers in a centralized and organized manner. In today's fast-paced digital environment, traditional job search and recruitment processes are often time-consuming, less transparent, and inefficient. This project aims to overcome these challenges by providing a modern, automated, and user-friendly system that simplifies the hiring process.

The system is intended to enable job seekers to create profiles, upload resumes, search for relevant job opportunities based on their skills, location, and experience, and apply for jobs online with ease. It also allows users to track the status of their applications and receive timely notifications regarding job updates. On the other hand, employers can post job openings, manage job listings, view applicant details, and shortlist or reject candidates efficiently through the platform.

Another important aim of the project is to ensure secure and reliable user authentication using technologies such as Spring Security and JWT, along with additional features like OTP verification to enhance data protection. The system also focuses on providing a responsive and interactive user interface using modern frontend technologies, ensuring accessibility across different devices.

1.3 Scope of the Project

The Job Portal Management System is designed to provide a comprehensive platform that facilitates interaction between job seekers and employers. The scope of this project covers the development of a web-based system that allows users to reg-

ister, create profiles, search for job opportunities, and apply for jobs online. It also enables employers to post job openings, manage job listings, and review applications efficiently.

The system includes features such as user authentication, role-based access control, resume upload, job search and filtering, and application tracking. It ensures secure communication using JWT authentication and OTP verification, enhancing data protection and user reliability. The platform is accessible through web browsers, making it convenient for users to access from different devices.

Additionally, the project scope extends to integrating advanced functionalities like email notifications, chatbot support, and map-based job location services, which improve user experience and system usability. The backend is developed using Spring Boot, ensuring scalability and performance, while MySQL is used for efficient data storage and management.

However, the scope is limited to a web-based application and does not include mobile app development or large-scale enterprise deployment. Future enhancements may include AI-based job recommendations, mobile applications, and integration with external job platforms.

In addition, the project incorporates advanced features like resume file upload, email notifications, chatbot support, and map-based job location services, which enhance the overall user experience. The system is designed using Spring Boot for backend development, Thymeleaf for frontend rendering, and MySQL for database management, ensuring scalability, maintainability, and efficient performance.

1.4 Frameworks and Technologies

The Job Portal Management System is developed using modern frameworks and technologies to ensure high performance, scalability, and security. The system is built using a full-stack approach that includes frontend, backend, and database technologies.

The backend of the application is developed using Spring Boot and Spring MVC, which provide a robust and scalable framework for handling server-side logic and RESTful API development. These frameworks support a layered architecture, making the system easy to maintain and extend. Spring Data JPA is used for database interaction and object-relational mapping, allowing efficient data handling and query execution.

The frontend is developed using HTML, CSS, and JavaScript, along with Thymeleaf as a template engine. Thymeleaf helps in dynamically rendering data from the backend to the user interface, ensuring a responsive and interactive user experience.

The system uses MySQL as the database for storing user details, job postings, and application records. It provides reliable and efficient data storage and retrieval.

For security, Spring Security is implemented to provide authentication and authorization, while JWT (JSON Web Token) is used for secure session management. Additional features such as OTP verification and email notifications are implemented using Java Mail services to enhance user security and communication.

Development tools such as IntelliJ IDEA / VS Code, Maven, and Postman are used for coding, dependency management, and API testing. The application runs on a local server (localhost), making it easy to test and deploy.

Spring Security is implemented to provide authentication and authorization, while JWT (JSON Web Token) is used for secure session management. Additional features

such as OTP verification and email notifications are implemented using Java Mail services to enhance user security and communication.

Overall, the combination of these frameworks and technologies ensures that the system is secure, scalable, efficient, and user-friendly, making it suitable for real-world applications.

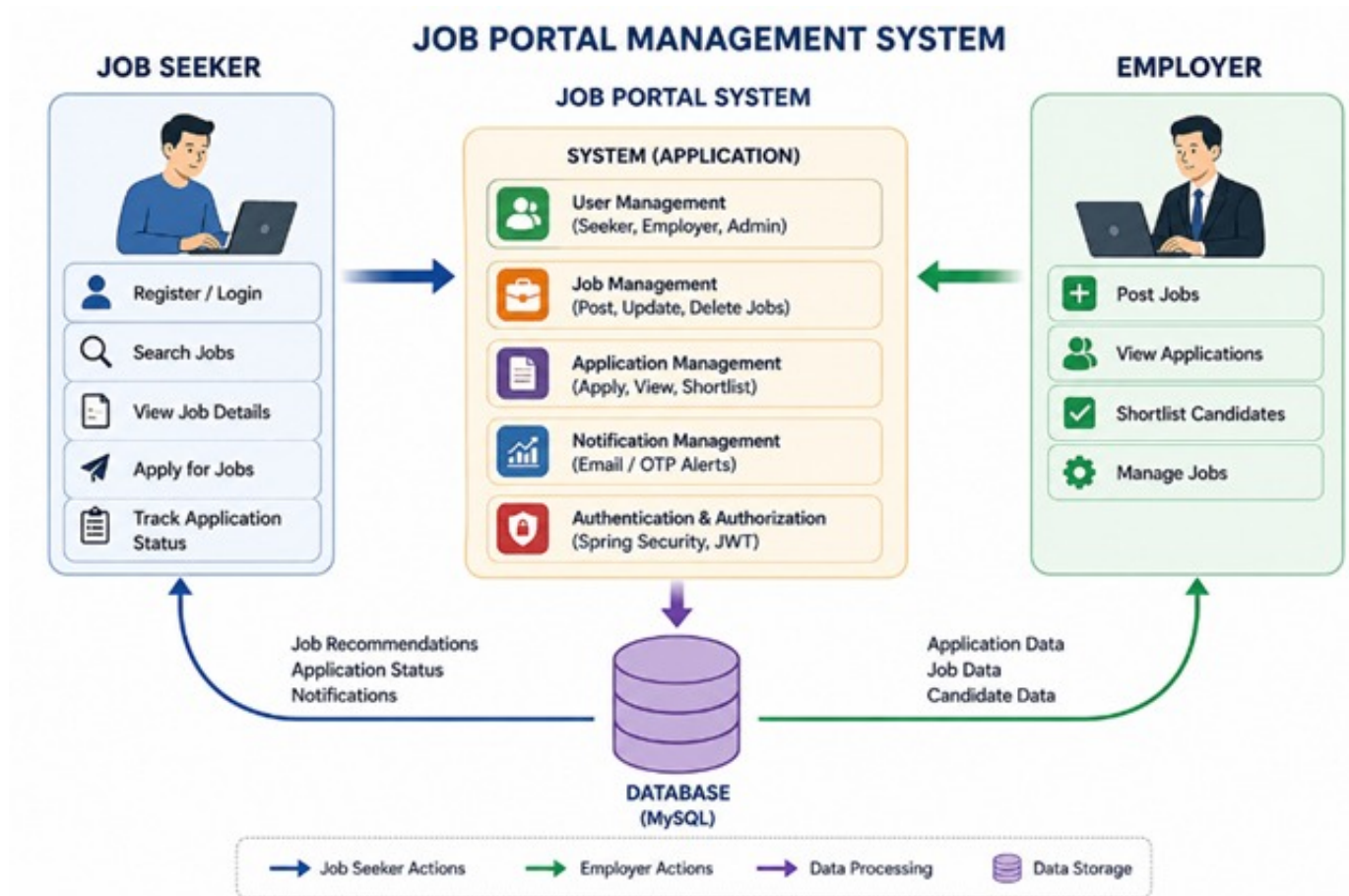


Figure 1.1: Framework

Figure 1.1 represents the Job Portal Management System architecture, showing interaction between job seekers, employers, and the system. Job seekers can register, search, and apply for jobs, while employers can post jobs and manage applicants. The system processes all data and stores it in a MySQL database, providing notifications and application updates.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

Before developing the **JOB PORTAL MANAGEMENT SYSTEM**, a detailed study of existing platforms related to career guidance, interview preparation, and community-based knowledge sharing was conducted. The purpose of this survey is to understand the strengths, limitations, and missing features in currently available platforms. Although many platforms provide job listings, interview reviews, or discussion forums, most of them do not offer a dedicated student-focused environment specifically designed for placement experience sharing. This chapter analyzes some of the most widely used platforms and identifies the research gap that motivated the development of this project.

2.1 Professional Networking Platforms

Professional networking platforms are widely used by students and job seekers to explore job opportunities, build professional connections, and learn about company recruitment processes.

2.1.1 LinkedIn

LinkedIn is one of the world's largest professional networking platforms, allowing users to create professional profiles, connect with recruiters, apply for jobs, and engage with industry-related content. The platform is widely used by students

and working professionals for career development and job discovery. As of 2026, LinkedIn has over 1.2 billion registered users globally. :contentReference[oaicite:0]index=0

Strengths:

- Provides direct access to recruiters and companies
- Offers professional networking opportunities
- Includes job listings, company pages, and career insights
- Supports skill endorsements and professional branding

Limitations:

- Placement experiences are not organized in a student-friendly format
- Interview discussions are scattered across posts and comments
- Limited focus on college-specific placement preparation
- Large number of applications can reduce visibility for students

Although LinkedIn is highly effective for professional networking, it does not provide a structured platform exclusively for sharing placement experiences and interview preparation knowledge.

2.2 Interview Review Platforms

Interview review platforms provide candidates with company-specific interview questions, recruitment experiences, and workplace insights.

2.2.1 Glassdoor

Glassdoor is a well-known platform where users can anonymously share interview experiences, salary reports, and company reviews. It is commonly used by job seekers to understand company culture and interview difficulty before attending recruitment processes.

Strengths:

- Provides interview questions from real candidates
- Offers salary and company review information
- Helps candidates understand recruitment patterns
- Supports anonymous community contributions

Limitations:

- Limited interaction between users
- Content is often focused on working professionals rather than students
- Some users question transparency and content moderation
- Search and filtering for academic placement preparation is limited

Community discussions on Reddit also show mixed user experiences regarding transparency and content moderation on Glassdoor.

Although Glassdoor provides useful interview insights, it does not create a collaborative student community where placement experiences can be actively discussed.

2.3 Community Discussion Platforms

Community discussion platforms enable users to share experiences, ask questions, and engage in topic-specific discussions.

2.3.1 Reddit

Reddit is a large community-driven discussion platform where users participate in topic-based communities known as subreddits. Many students and professionals use Reddit to discuss interview experiences, coding challenges, and job preparation strategies. Reddit had over 121 million daily active users in 2025.

Strengths:

- Strong community engagement
- Open discussions and honest opinions
- Voting system helps useful content gain visibility
- Covers a wide range of technical and career topics

Limitations:

- Information is spread across multiple communities
- No structured organization for placement experiences
- Difficult to filter company-specific student experiences
- Content quality may vary across discussions

Reddit inspired the community-based discussion and upvoting model implemented in the Placement Experience Platform.

2.4 Educational and Coding Platforms

Educational platforms help students prepare technical concepts, coding challenges, and interview-related topics.

2.4.1 GeeksforGeeks and Similar Platforms

Platforms such as **GeeksforGeeks**, coding forums, and interview preparation portals provide technical articles, coding practice problems, and interview preparation materials.

Strengths:

- Provides technical learning resources
- Helps improve problem-solving skills
- Covers data structures, algorithms, and coding interviews
- Useful for technical preparation

Limitations:

- Does not provide real student placement stories
- Limited community-driven placement discussions
- Does not support institution-specific placement insights

Although these platforms support technical preparation, they do not capture actual placement journeys and company-specific experiences from students.

2.5 Gap Analysis and Motivation

After studying existing platforms such as LinkedIn, Glassdoor, Reddit, and technical learning communities, several research gaps were identified. Most existing systems either focus on professional networking, interview reviews, or technical preparation independently. Very few platforms combine all these features in a student-focused environment.

The major gaps identified are:

- Lack of a centralized platform exclusively for student placement experiences
- Absence of structured company-wise interview experience sharing
- Limited peer-to-peer academic collaboration
- Lack of integrated features such as comments, upvotes, bookmarks, and company following
- Difficulty in finding authentic, organized, and college-relevant placement insights

To overcome these limitations, the job portal management system is developed as a dedicated community-driven system where students can share authentic placement experiences, interact with peers, and access organized company-specific preparation resources.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

The frontend of the job portal management system is designed to provide an interactive, responsive, and user-friendly interface for students. It allows users to perform activities such as registration, login, profile management, post creation, viewing placement experiences, and interacting with other users through comments and upvotes. The frontend is developed using HTML5, CSS3, and JavaScript, which provide flexibility in designing modern web interfaces and handling dynamic content.

3.1.1 Environment Setup

To develop and run the frontend application, a suitable development environment is configured with the required tools and dependencies. The user interface is built using standard web technologies, and the source code is managed using code editors such as Visual Studio Code. Modern web browsers such as Google Chrome are used for testing and debugging the application. JavaScript is used to handle API communication, user interactions, and asynchronous operations. Browser developer tools are used during testing to inspect layouts, debug scripts, and monitor API requests.

3.1.2 Structure of the Project

The frontend project is organized in a modular structure to improve maintainability and scalability. The project contains separate HTML files for pages such as login, signup, profile management, company pages, and placement post views. CSS files are used to maintain consistent styling across the platform, while JavaScript files handle dynamic interactions and API communication. A custom `api.js` file

is implemented to simplify communication with backend REST APIs. This modular structure allows easy updates, better code organization, and separation of design, logic, and content.

3.1.3 Execution Procedure

To execute the frontend application, all frontend files are loaded through a web browser or a local development server. The user interface sends requests to the backend APIs whenever actions such as login, registration, post creation, commenting, or profile updates are performed. JavaScript processes user inputs, validates form data, sends API requests, and dynamically updates content based on server responses. During execution, browser developer tools are used to monitor network requests, debug JavaScript functions, and verify API communication.

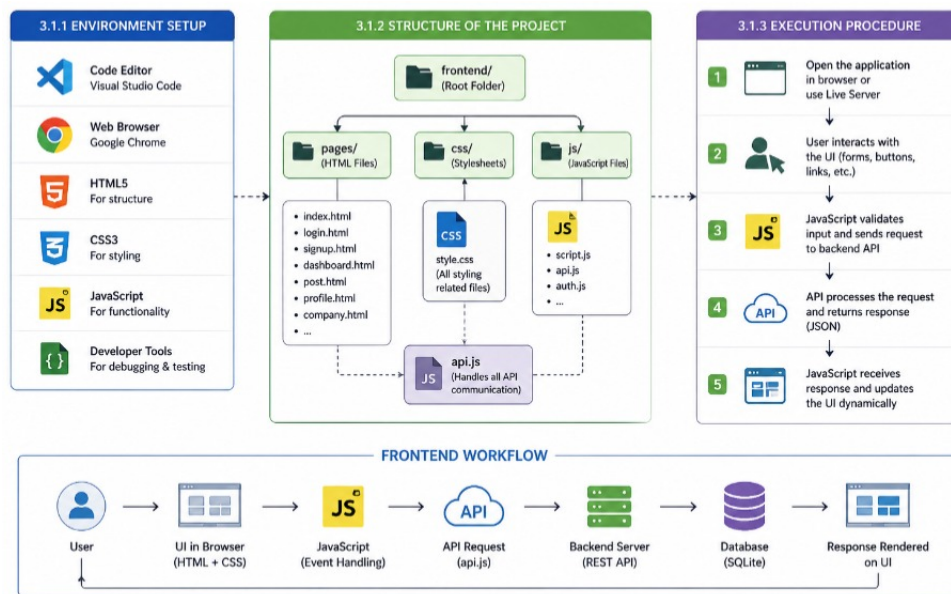


Figure 3.1: Frontend implementation

Figure 3.1 shows the frontend implementation structure and workflow of the Job Portal system. It explains how user actions in the browser are processed using JavaScript and APIs to communicate with the backend.

3.2 Backend Implementation

The backend of the job portal management system is responsible for handling business logic, database operations, authentication, security, and API communication. It is developed using Java and Spring Boot, which provide a scalable and modular architecture for building enterprise-level applications. The backend manages user authentication, placement posts, comments, notifications, bookmarks, upvotes, and company-related operations.

3.2.1 Environment Setup

The backend development environment requires Java Development Kit (JDK 17), Maven, and a suitable integrated development environment such as IntelliJ IDEA. Spring Boot dependencies are managed using Maven, which automatically downloads and configures required libraries. SQLite is configured as the database management system, and application properties are defined to establish database connections and server configurations. Tools such as Postman are used to test API endpoints and validate backend functionality during development.

3.2.2 Structure of the Project

The backend project follows a layered architecture under the package structure `com.predecessor`. The controller layer handles incoming HTTP requests and API endpoints. The service layer contains business logic and data processing operations. The repository layer manages database communication using JPA interfaces. The entity layer represents database tables as Java classes, while DTO classes are used for data transfer between layers. The security layer contains JWT utilities, authentication filters, and security configurations to protect application resources. This architecture ensures clean code organization, reusability, and easy maintenance.

3.2.3 Execution Procedure

To execute the backend application, the project is first built using Maven, which downloads dependencies and compiles the source code. After successful compilation, the Spring Boot application is started using the embedded server. Once the server starts, REST API endpoints become accessible through the configured port. Frontend requests are then processed by the backend controllers, validated by service classes, and stored in the SQLite database through repository interfaces. JWT authentication is applied to protected routes, ensuring secure communication between users and the system.

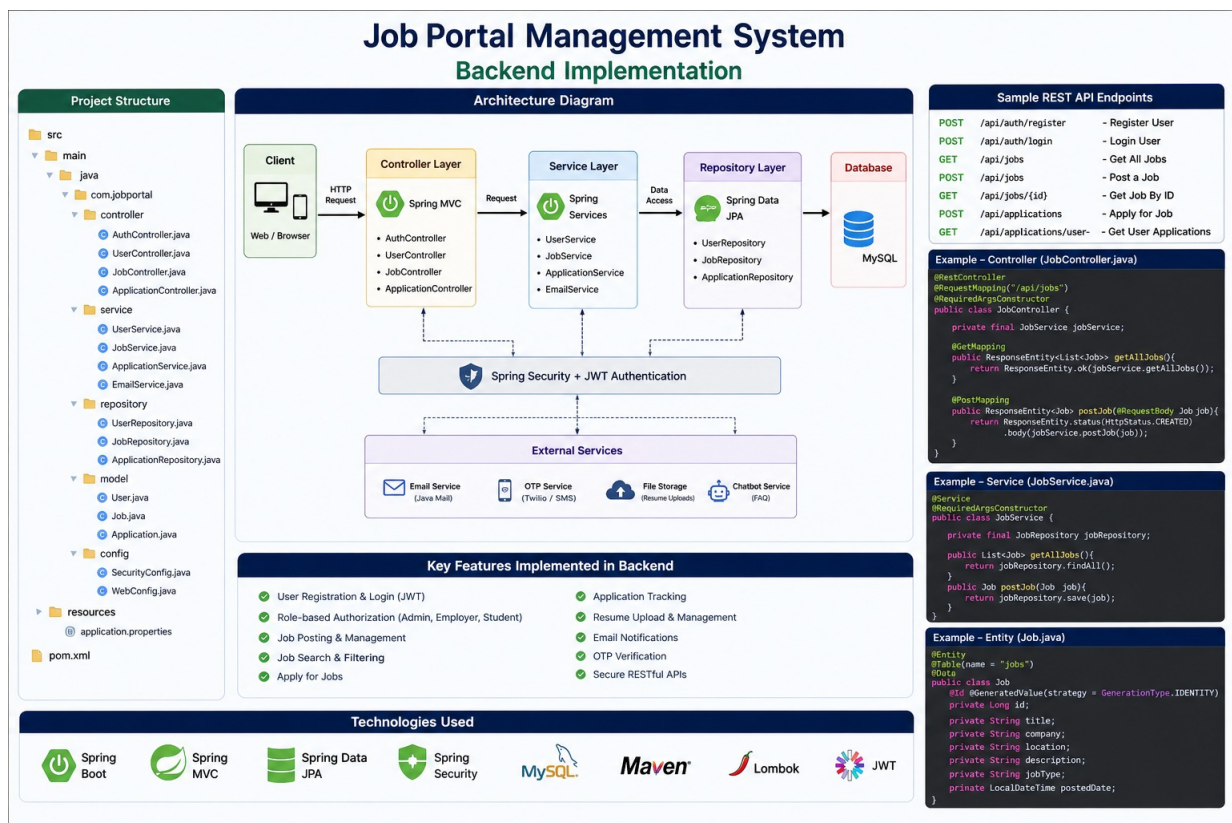


Figure 3.2: Backend implementation

Figure 3.2 shows the backend implementation and architecture of the Job Portal Management System. It explains how the controller, service, and repository layers interact with the MySQL database.

3.3 Database Implementation

The database layer of the job portal management system is responsible for storing, managing, and retrieving all application data in an organized and efficient manner. The platform uses SQLite as the database management system because it is lightweight, serverless, easy to configure, and suitable for academic and small to medium-scale applications. The database stores critical information such as user profiles, placement posts, interview rounds, comments, bookmarks, notifications, upvotes, and company-related data. To simplify database interaction, is used as the Object Relational Mapping (ORM) framework, while is used as the persistence standard for managing entity relationships and database operations.

3.3.1 Database Configuration

The database configuration of the project is managed through the Spring Boot application configuration files. SQLite is connected to the backend using the JDBC driver, which enables communication between the application and the database file. Hibernate and JPA are configured to automatically map Java entity classes to database tables and manage schema updates during development. The database file used in the project is `predecessor.db`, which stores all application data in a single portable file. Spring Boot automatically initializes the database connection during application startup, ensuring smooth communication between the backend services and the database layer.

3.3.2 Schema Structure

The database schema is designed using a relational model to maintain data consistency and reduce redundancy. Each major component of the application is represented as a separate entity, and relationships are established using primary keys and foreign keys. For example, users can create multiple posts, posts can contain mul-

multiple comments, and users can interact with posts through upvotes and bookmarks. Additional relationships are maintained for notifications, company following, and interview rounds. This structured schema ensures efficient data retrieval, proper normalization, and easy maintenance of application data.

3.3.3 Tables Created

The core functionalities of the Placement Experience Platform are supported by multiple database tables. The `users` table stores user account details such as username, email, encrypted password, and profile information. The `posts` table stores placement experiences, company names, interview details, and preparation tips shared by users. The `comments` table manages user discussions under each post. The `upvotes` table stores voting interactions to prevent duplicate votes and track engagement. The `bookmarks` table stores saved posts for future reference. The `notifications` table manages user activity updates, while the `company_follow` table stores company subscription details. Additionally, the `rounds` table is used to store multiple interview rounds such as aptitude, technical, coding, and HR rounds associated with placement experiences. These tables collectively support the complete functionality of the platform.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The job portal management system follows a layered full-stack architecture designed to ensure modularity, scalability, and secure communication between system components. Each layer performs a specific function and works together to provide smooth system operation.

Main Layers of the Architecture:

- **Presentation Layer:** Developed using HTML5, CSS3, and JavaScript, this layer provides the user interface for login, profile management, and placement posts.
- **Application Layer:** Built using Spring Boot, this layer handles API requests and manages communication between frontend and backend services.
- **Business Logic Layer:** This layer processes core functionalities such as authentication, post management, comments, and notifications.
- **Database Layer:** SQLite is used for data storage, while Hibernate and JPA manage database operations and entity relationships.

Security is implemented using Spring Security and JWT authentication to protect user data and system resources.

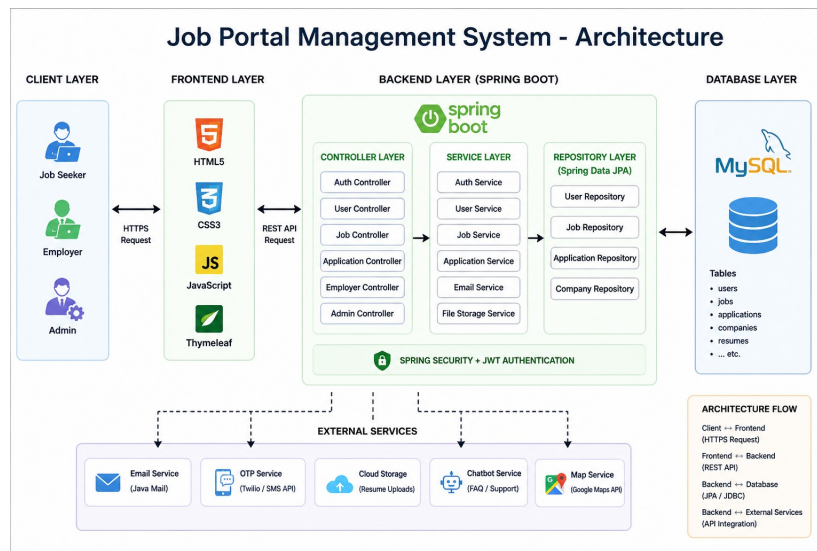


Figure 4.1: Architecture

Figure 4.1 the overall architecture of the Job Portal Management System, including client, frontend, backend, and database layers. It explains how requests flow from users through Spring Boot components to the MySQL database, with security and external services integrated.

4.2 DataFlow Diagram

The Data Flow Diagram (DFD) shows how data moves through the Placement Experience Platform and how user requests are processed by different modules.

Data Flow Process:

- Users perform actions such as registration, login, post creation, or commenting through the frontend interface.
- The frontend sends requests to the backend using REST APIs.
- Backend controllers process requests through service classes.
- Required data is stored or retrieved from the SQLite database.

- The processed response is sent back to the frontend and displayed to the user.

This flow ensures efficient communication and smooth system performance.

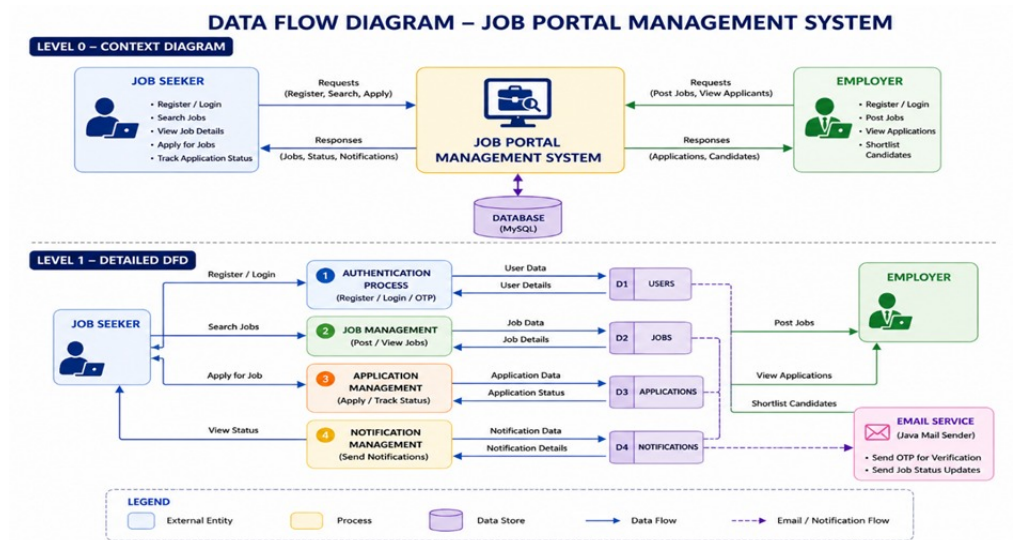


Figure 4.2: Dataflow

Figure 4.2 represents the Data Flow Diagram (DFD) of the Job Portal ManagementSystem showing interactions between job seekers, employers, and the system.

4.3 Module 1: User Authentication and Profile Management

This module acts as the entry point of the system and is responsible for managing user registration, login, authentication, and profile operations.

Functions of this module:

- User registration with email and password
- Secure login and token generation
- Password encryption and validation
- Profile creation and management
- Access control for authenticated users

Implementation: This module is implemented using Spring Security and JWT authentication. After successful login, a JWT token is generated and used for accessing protected resources.

4.4 Module 2: Placement Post and Community Interaction

This module handles placement experience sharing and community engagement within the platform. It allows students to share interview experiences and interact with other users.

Functions of this module:

- Creating placement experience posts
- Editing and deleting posts
- Adding interview rounds and preparation tips

- Commenting on shared posts
- Upvoting useful content
- Bookmarking posts for future reference

Implementation: This module is implemented using REST APIs, controller classes, and database entities for managing posts and interactions.

4.5 Module 3: Company Tracking and Notification Management

This module manages company-related updates and user notifications. It helps students stay informed about placement activities related to specific companies.

Functions of this module:

- Following companies of interest
- Receiving company-related updates
- Notification generation for interactions
- Tracking comments, upvotes, and bookmarks

Implementation: This module uses notification services, company entities, and database relationships to maintain user engagement and activity tracking.

Advantages of this Project

The Placement Experience Platform offers multiple benefits for students preparing for placements.

a) Centralized Information Sharing: All placement experiences, interview questions, and preparation strategies are available in one organized platform, reducing the time spent searching across multiple sources.

b) Secure and Reliable Access: The platform uses Spring Security and JWT authentication to ensure secure access, protect user data, and maintain system reliability.

c) Community-Based Learning: Features such as comments, upvotes, bookmarks, and company following encourage collaboration and help students learn from real placement experiences.

d) Efficient Application Tracking: Efficient Application Tracking Job seekers can track the status of their applications in real-time, improving transparency in the recruitment process.

e) Advanced Features Integration: Features like chatbot support, OTP verification, and map integration enhance user experience and system functionality.

f) Employer Management Features: Automated processes such as notifications, filtering, and data management reduce manual effort and save time for both users and employers.

g) User-Friendly Interface: The system provides a simple and responsive UI, making it easy for users to navigate and use the platform effectively.

h) Scalable and Flexible Architecture: Built using Spring Boot and MySQL, the system is scalable and can be extended with additional features in the future.

Disadvantages

Although the project offers several advantages, certain limitations exist.

- The platform depends on active user participation for content generation.
- SQLite may face scalability limitations for large-scale deployments.
- Advanced features such as real-time chat and mobile support are not included in the current version.

Chapter 5

RESULTS AND DISCUSSIONS

This chapter presents the implementation results of the job portal management system and discusses the performance of different modules developed during the project. The system was successfully designed and implemented as a full-stack web application that provides students with a centralized platform for sharing placement experiences, interview insights, and preparation strategies. The platform was tested for functionality, usability, security, and database operations to ensure smooth performance across different modules.

5.1 System Implementation Results

The developed system successfully integrates frontend interfaces, backend APIs, authentication mechanisms, database operations, and user interaction features. All major functionalities including user registration, login, profile management, placement post creation, commenting, bookmarking, upvoting, company following, and notification management were tested and verified.

The platform allows students to securely create accounts, log in, and share placement experiences with structured interview rounds. Other users can interact with these posts through comments and upvotes, creating an active learning environment. The system also successfully handles secure API communication using JWT authentication and stores all application data efficiently using SQLite.

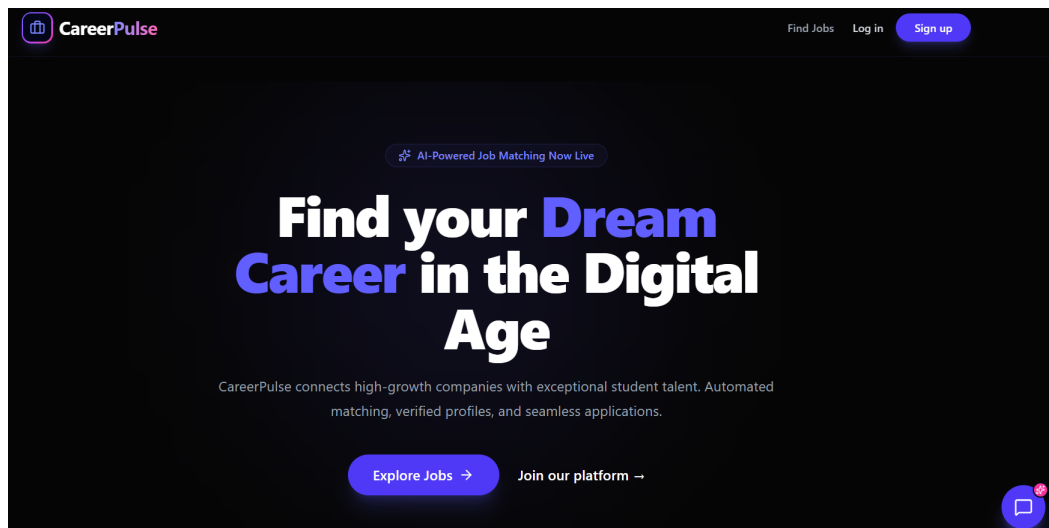
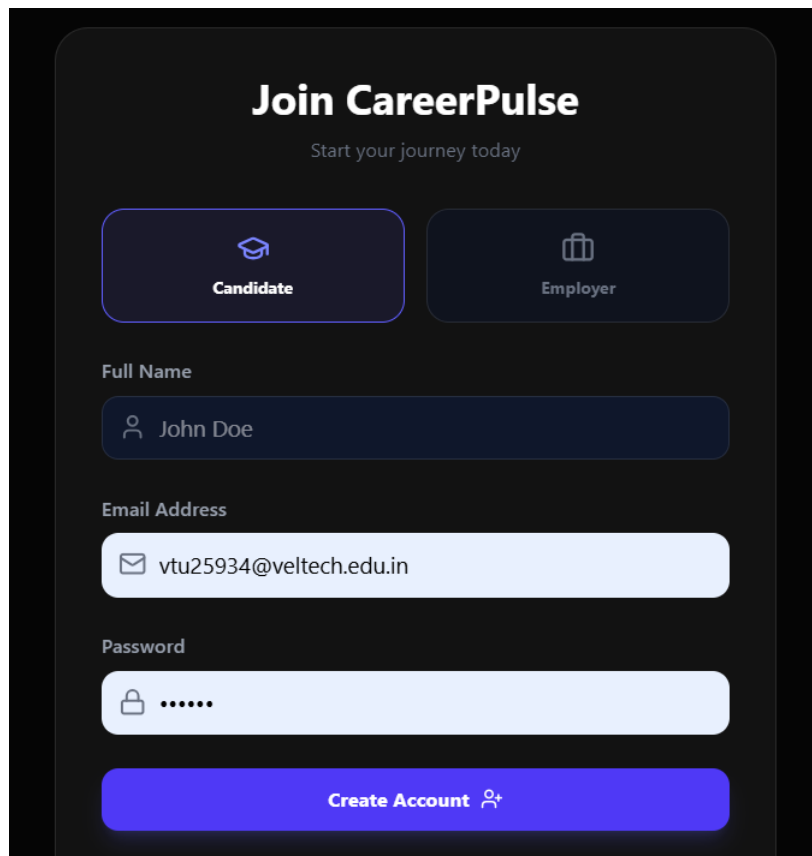


Figure 5.1: Homepage

Homepage:

Figure 5.1 shows the homepage of the job portal management system. This page acts as the main dashboard of the application, where users can view placement experiences shared by other students. Each post contains important details such as company name, job role, interview status, and experience summary. The homepage also provides quick navigation options and allows users to explore placement discussions, interact with posts, and access different modules of the system.



The image shows a dark-themed user login page for 'CareerPulse'. At the top, the title 'Join CareerPulse' is displayed in white, with the subtitle 'Start your journey today' below it. Two buttons are present: 'Candidate' with a graduation cap icon and 'Employer' with a briefcase icon. Below these are three input fields: 'Full Name' with the value 'John Doe', 'Email Address' with the value 'vtu25934@veltech.edu.in', and 'Password' with masked characters '.....'. A blue 'Create Account' button with a user icon is at the bottom.

Figure 5.2: User login page

User Login Page:

Figure 5.2 shows the user login page of the system. This page allows registered users to securely access the platform by entering their email address and password. User credentials are validated through the backend authentication system, and a JWT token is generated after successful login. This authentication mechanism ensures secure access to protected resources and prevents unauthorized users from accessing system functionalities.

The screenshot shows a dark-themed web form titled "Post an Opportunity" with the subtitle "Reach thousands of talented students instantly." The form includes several input fields: "Job Title" with a placeholder "e.g. Senior Frontend Developer", "Category" with a dropdown menu showing "Engineering", "Location" with a placeholder "e.g. New York (Remote)", "Experience Level" with a dropdown menu showing "Fresher", and "Offered Salary" with a placeholder "e.g. ₹5,00,000 - ₹8,00,000 / year". Below these is a large text area for "Job Description" with a placeholder "Describe roles, responsibilities, and requirements...". At the bottom left, there is a small icon and text stating "AI will automatically tag this job for optimal search matching." At the bottom right, there is a "Post Job" button with a right-pointing arrow.

Figure 5.3: Post details page

Post Details Page:

Figure 5.3 shows the post details page of the job portal management system. This page displays complete information about a placement experience shared by a user, including the company name, job role, interview rounds, and overall experience summary. Users can read detailed interview insights, technical questions, and preparation strategies shared by other students. The page also supports interactive features such as commenting, upvoting, and bookmarking, which encourage community participation and help students identify useful placement content.

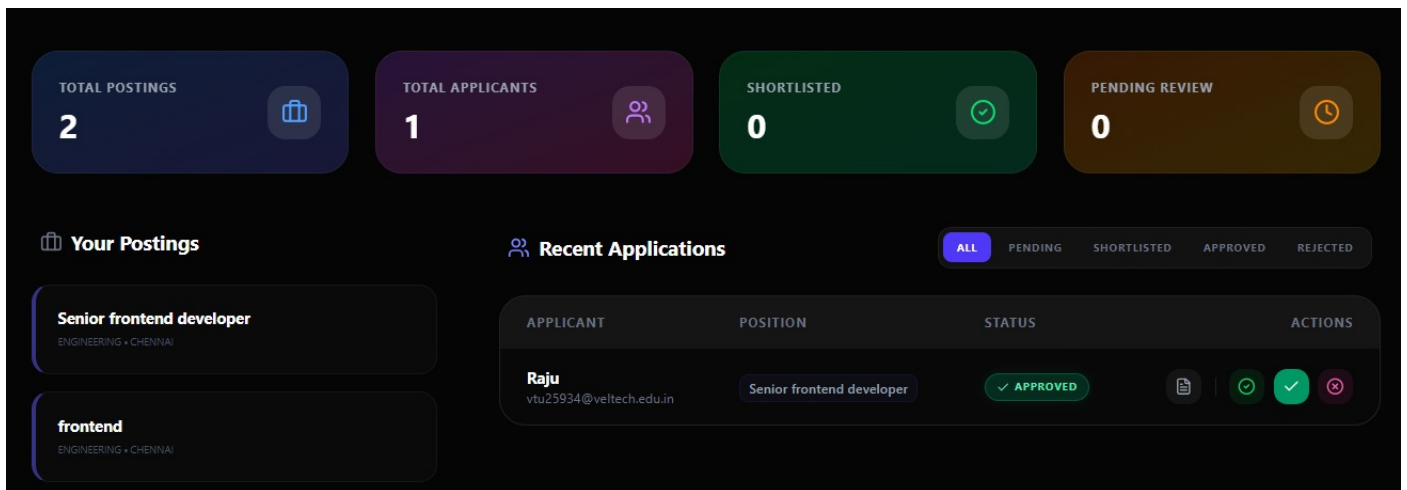


Figure 5.4: Posting page

Posting Page:

Figure 5.4 shows the posting page of the system. This page allows authenticated users to share their placement experiences by entering details such as company name, job role, placement type, interview year, and round-wise experiences. Users can add technical questions, coding challenges, and preparation tips based on their interview journey. This feature helps build a structured knowledge base where students can contribute authentic placement insights and support others in their preparation.

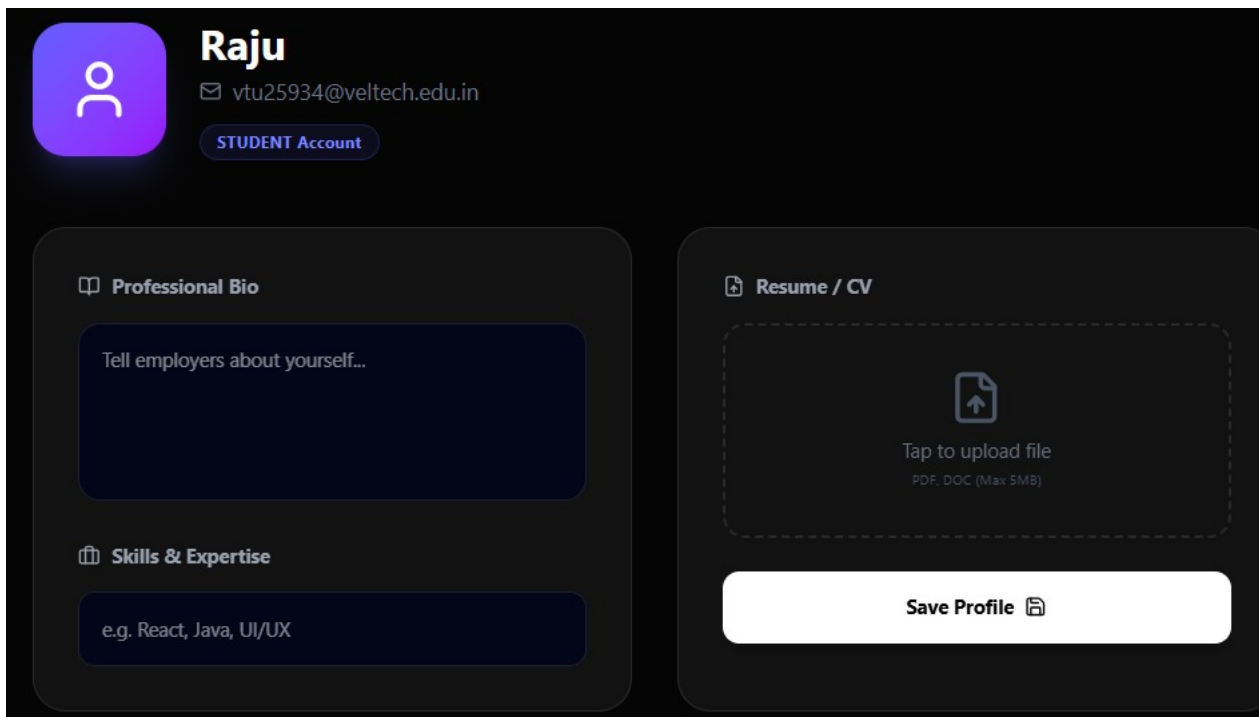


Figure 5.5: User profile page

User Profile Page:

Figure 5.5 shows the user profile page of the job portal management system. This page displays user information such as profile details, shared placement experiences, and user activity within the platform. It allows users to manage their personal information and view their contributions. The profile page helps users maintain their placement history and track their engagement within the community.

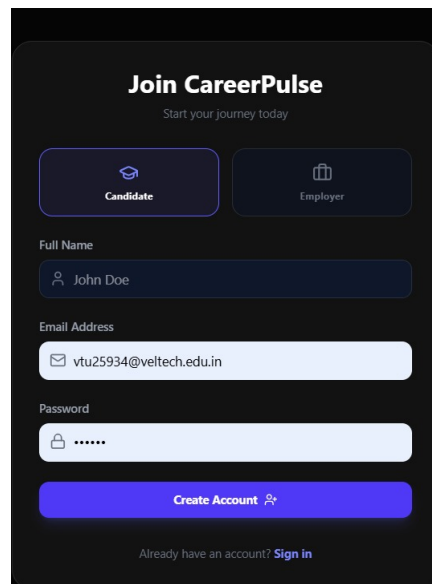
The image shows a mobile app interface for 'Join CareerPulse'. At the top, the title 'Join CareerPulse' is displayed in white, with the subtitle 'Start your journey today' below it. There are two buttons: 'Candidate' with a graduation cap icon and 'Employer' with a briefcase icon. Below these are three input fields: 'Full Name' with the text 'John Doe', 'Email Address' with the text 'vtu25934@veltech.edu.in', and 'Password' with masked characters '*****'. A large blue button labeled 'Create Account' with a user icon is at the bottom. At the very bottom, there is a link that says 'Already have an account? Sign in'.

Figure 5.6: User signup page

User Signup Page:

Figure 5.6 shows the user signup page of the system. This page allows new users to create an account by entering their personal details such as name, email address, and password. The entered data is validated before account creation to ensure data accuracy and security. After successful registration, users can access the platform and participate in placement-related discussions and knowledge sharing.



Figure 5.7: Application page

Application Page:

Figure 5.7 shows the “Your Applications” dashboard in the Job Portal Management System. It allows users to track the status of job applications in real time. The interface displays job details such as job title, location, and application date, along with status indicators like Approved, Pending, Shortlisted, or Rejected. Users can filter applications using tabs (All, Pending, Shortlisted, Approved, Rejected) for better organization. The design provides a clear and user-friendly way to monitor application progress.

5.2 Module Testing Results

Each module of the system was tested individually to ensure proper functionality and integration. The testing results show that all major modules performed successfully without critical errors. The system maintained stable communication between frontend and backend components throughout testing.

Table 5.1: System Testing and Functional Analysis

S.No	Module Name	Functionality Tested	Status
1	User Authentication	User registration, secure login, password validation, and JWT token generation were tested successfully.	Passed
2	Profile Management	User profile creation, updating personal details, and profile information retrieval were verified successfully.	Passed
3	Post Management	Creation, editing, viewing, and deletion of placement experience posts were tested successfully.	Passed
4	Comment System	Adding comments, viewing discussions, and managing post interactions were tested.	Passed
5	Upvote System	Upvoting posts, removing votes, and dynamic vote count updates were verified.	Passed
6	Bookmark Module	Saving placement posts and retrieving bookmarked content were tested successfully.	Passed
7	Notification Module	Notification generation for comments, votes, and user activities was verified.	Passed
8	Database Operations	Data insertion, retrieval, update, and deletion operations were executed successfully.	Passed

5.3 Performance Analysis

The performance of the application was evaluated based on response time, database operations, and user interaction efficiency. The frontend interfaces loaded successfully, and API communication between frontend and backend was executed without significant delay. User authentication requests, post creation requests, and comment updates were processed efficiently.

The SQLite database handled CRUD operations successfully, including data insertion, updates, retrieval, and deletion. Hibernate and JPA managed database relationships effectively, reducing manual query complexity and improving overall system performance.

5.4 Discussion

The implementation of the Placement Experience Platform demonstrates that a centralized student-focused placement system can significantly improve access to authentic placement information. Features such as post creation, community interaction, bookmarking, and company tracking help students prepare more effectively compared to traditional scattered information sources.

The integration of Spring Boot, JWT authentication, SQLite, and frontend technologies resulted in a secure and reliable platform. Community features such as comments and upvotes encourage peer-to-peer learning and increase user engagement. The modular architecture also makes the system easy to maintain and extend in future versions. Future improvements such as real-time messaging, mobile support, and AI-based recommendations can further enhance the overall user experience.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The job portal management system was successfully designed and implemented as a full-stack web application that provides students with a centralized platform for sharing and accessing placement-related experiences. The system effectively addresses the problem of scattered placement information by organizing interview experiences, company-specific insights, technical questions, and preparation strategies in a structured and secure environment.

The platform successfully integrates essential features such as user authentication, profile management, placement post creation, commenting, bookmarking, upvoting, company following, and notification management. The use of Java, Spring Boot, SQLite, and modern frontend technologies ensured secure communication, efficient data management, and a smooth user experience. Through community-driven interaction, the platform encourages peer-to-peer learning and helps students prepare more confidently for campus recruitment processes.

Overall, the project achieved its primary objectives by creating a reliable, scalable, and user-friendly placement knowledge-sharing system that can significantly improve placement preparation and student engagement.

6.2 Future Enhancements

Although the current implementation successfully meets the core requirements, several enhancements can be made in future versions to improve functionality and user experience.

- Development of a mobile application for easier access across devices.
- Integration of real-time messaging and live discussion features using WebSockets.
- Implementation of AI-based company recommendations and personalized preparation suggestions.
- Addition of analytics dashboards to track user engagement and popular placement trends.
- Integration of resume analysis and mock interview support for better career preparation.

These future enhancements can make the platform more interactive, scalable, and beneficial for a larger student community.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Wks	Justification
WP1	Depth of Knowledge	Requires in-depth engineering knowledge	WK3, WK4, WK5, WK6	Requires knowledge in Java, Spring Boot, REST APIs, frontend technologies, and security implementation for building a scalable placement platform.
WP2	Conflicting Requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances ease of use for students while maintaining secure authentication, data validation, and system reliability.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	Involves system design, database schema creation, API development, validation handling, and efficient user interaction mechanisms.
WP4	Familiarity	Partially new problems	WK4, WK8	Integrating placement experience sharing, company tracking, and community interaction introduces new design challenges.
WP5	Applicable Codes	Limited standard solutions	WK6	Requires custom implementation for post interactions, bookmarking, and company-follow features.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Addresses the requirements of students, contributors, and administrators to ensure smooth knowledge sharing.
WP7	Interdependence	System level integration	WK3, WK5, WK6	Integrates frontend UI, backend services, database, security, and notification modules into one unified platform.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes Java, Spring Boot, Spring Security, JWT, Hibernate, SQLite, HTML, CSS, and JavaScript for full-stack development.
EA2	Level of Interactions	Complex interactions between modules	Handles interactions between authentication, post management, comments, bookmarks, notifications, and company tracking modules.
EA3	Innovation	Application of modern technologies	Implements a centralized placement experience platform with secure APIs, user engagement features, and scalable architecture.
EA4	Consequences to Society	Impact on users and society	Improves student placement preparation, increases accessibility to real experiences, and reduces dependency on scattered information sources.
EA5	Familiarity	Beyond standard practices	Combines community learning, secure authentication, and structured placement insights into a single digital platform.

7.3 SDG Mapping

Table 7.3: SDG Mapping for the Project

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Provides students with placement preparation resources, interview experiences, and peer learning opportunities beyond classroom education.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital innovation by creating a centralized platform for career guidance and placement knowledge sharing.
SDG 10	Reduced Inequalities	Helps all students access equal placement opportunities and preparation resources regardless of their social network or connections.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent platform access through authentication, authorization, and protected data handling.
SDG 17	Partnerships for the Goals	Encourages collaboration, knowledge sharing, and peer-to-peer learning among students and academic communities.

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